

ProbOS — Month 3 Plan

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Status

Per `docs/monthly_plans/overall/main.tex`, Month 3 was always DIRECTIONAL rather than pre-specified in detail – Month 1’s retrospective explicitly flagged over-specifying future months as a mistake, corrected by deferring detailed week-by-week planning until the month is actually about to begin. This document intentionally states THEMES, not a committed day-by-day schedule; the latter will be written (following the same plan-then-retrospective process used for every week so far) once Month 3’s actual work begins.

Candidate themes, in rough priority order

GPU kernel path

Referenced as a genuine open item in `docs/sbir_phase1_onepager.md`’s SBIR Ask (“GPU kernel, 10x speedup over CPU, building on the existing C++/OpenMP path”). The Month 1 Week 4 C++/OpenMP kernel achieved a measured 7x speedup over the Python engine; a GPU path would need its own honest profiling discipline (the same “measure, don’t assume” standard that found DP’s real 1040x/45x wins alongside one genuine non-win in Month 1 Week 4) before any speedup number is claimed.

Extend the REST API’s model registry

Also an open SBIR ask: `python/server/main.py` currently only registers `BatteryModel2Cell`. Extending it to the three Week 4 cross-discipline models (option pricer, ED queue, clinical trial) is low-risk, since the pattern is already proven for one model and the models themselves are unchanged.

PDSL v0.2 – control flow

Month 1 Week 4’s PDSL compiler v0.1 supports only flat drift/diffusion declarations, no loops or conditionals. Genuinely useful, but not urgent – no current model has needed it yet.

Apply the particle filter to real (non-synthetic) data

The most consequential open SBIR ask (“apply the particle filter to real sensor data from a pilot partner”), and the one most dependent on external factors (an actual pilot partnership) rather than purely technical work. Cannot be scheduled with certainty; flagged here so it is not forgotten.

Continued quality-standard maintenance

Per the mid-Month-2 hardening pass: `pre_week_audit.sh` and `docs/standards/quality_standards.md` are living documents. Any new gap discovered (declared-but-unenforced checks, stale documentation, vestigial code patterns) should be closed in the same session it is found, not deferred – the standing practice established this month, to be continued.

Process

As with every prior month, Month 3's actual week-by-week plan will be written and compiled to PDF (`docs/monthly_plans/month3/weekN/main.tex`) before that week's work begins, with a retrospective appended once it completes. This document will be revisited and made more concrete once Month 3 actually starts.